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EDUCATION

Higher Education (Software Engineering Bachelor's Degree)

Universidad Autónoma de Yucatán | Facultad de Matemáticas

August 2022 - December 2027, Mérida, Yucatán, México

Relevant Coursework: Data Structures, Algorithms, App/Web Development, Software Construction, Discrete Mathematics, Superior Algebra.

SKILLS

Spoken Languages: Spanish (Native), English (C2 Proficient), French (Conversational).

Programming Languages: Python, Java, C++, C, JavaScript

Tools: Git, Docker, Linux, AWS, Spreadsheets, SQL

Parallel/HPC: OpenMP, CUDA, MPI, POSIX Threads.

AWARDS AND COMPETITIONS

ICPC National/Latin American Finalist

Universidad Autónoma de Yucatán | Facultad de Matemáticas

May 2025 - September 2025, Mérida, Yucatán, México

Universidad Jesuita de Guadalajara

November 2025, Tlaquepaque, Jalisco, México

- Qualified and competed at the ICPC National/Latin American finals, placing 76th nationwide and 299th in Latin America among hundreds of highly skilled teams.
- Solved complex algorithmic problems under time pressure, applying several data structures and dynamic programming techniques.

ANIEI 2024 National Programming Contest

October 2024, Mérida, Yucatán, México

Universidad Autónoma de Yucatán | Facultad de Matemáticas

- Secured 15th place out of numerous teams at the national level.

PROJECTS

AWS-hosted Student Management REST API

Universidad Autónoma de Yucatán | Facultad de Matemáticas

August 2025 - December 2025, Mérida, Yucatán, México

- Developed a FastAPI REST app on EC2 with persistent storage on RDS via the SQLAlchemy ORM, and session endpoints backed by DynamoDB tables.
- Built a profile picture upload flow to a public S3 bucket via the AWS SDK for Python.
- Added automated email notifications via Simple Notification Service (SNS) topics.

Parallel Image Filtering via Matrix Convolution

Universidad Autónoma de Yucatán | Facultad de Matemáticas

August 2025 - December 2025, Mérida, Yucatán, México

- Achieved 3.25x CPU speedup using OpenMP with "collapse" and "schedule" directives, and implemented an MPI version to document communication overhead trade-offs.
- Generated grayscale outputs using FreeImage across OpenMP, CUDA, and MPI, by implementing a 3x3 matrix-convolution image filtering pipeline in C++.
- Reduced average runtime by 9.38x by mapping pixels to CUDA threads under benchmarked block sizes from 128 to 1024 threads per block.

Human Activity Recognition and Postural Transitions Based on Data Fusion with Python

Universidad Autónoma de Yucatán | Facultad de Matemáticas

October 2023 - December 2023, Mérida, Yucatán, México

- Developed Machine Learning Pipelines for Activity Recognition and Postural Transitions reading triaxial sensor data and processing it with Pandas, NumPy, and Scikit-Learn.
- Designed, analyzed and optimized ensemble models with the help of dimensionality reduction as well as grid search and cross validation to optimize multiclass classification.
- Tested Logistic Regression, Random Forest, and Perceptron classifiers, applying Borderline SMOTE for class imbalance and plotting results through Seaborn and Matplotlib.

EXPERIENCE

Lead researcher

Texas A&M University

May 2024 - August 2024, College Station, Texas, United States

- Led a group of 7 students from both Texas A&M University and UADY to develop a research proposal addressing the reliability and practicality of electric vehicles versus gas powered vehicles in Mérida (Yucatán) and Houston (Texas).
- Coordinated in a hybrid team environment to meet short-term and long-term research objectives, ensuring project alignment with local needs.
- Presented our final project to several authorities, including the director of the Faculty of Mathematics and the rector of the Universidad Autónoma de Yucatán (UADY).

Support Manager

Universidad Autónoma de Yucatán | Facultad de Matemáticas

October 2023 - December 2023, Mérida, Yucatán, México

- Assisted in the successful launch and management of a new e-learning platform, contributing to its stability and user adoption by over 200 students.
- Managed and optimized databases to ensure efficient data handling and massive user registration.
- Provided customized 1:1 support to students, addressing technical issues and improving their learning experience.

CERTIFICATIONS AND TRAINING

AWS Certified Cloud Practitioner - Amazon Web Services

AWS re/Start Cohort Program Graduate - Amazon Web Services

English Certificate (C2 Proficient) - EF SET

Data Analytics Professional Certificate - Google

Endpoint Security training - Cisco

Introduction to Cybersecurity training - Cisco